

Discrete Geodesic Topology and Conservative Finite-Volume Spatial Fluxes: Metric-Preserving Unit Vector Equivalence in Planetary Biogeochemical Grids

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Official Project Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

Sprint 070 — Biogeochemical Transport & Spatial Adjacency Architecture

Abstract

Planetary biogeochemical simulations operating on discrete partitions of the two-sphere $\mathbb{S}^2$ rely fundamentally on exact boundary geometry to enforce physical conservation laws. In Discrete Global Grid Systems (DGGS), such as the icosahedral hexagonal hierarchy of H3, floating-point round-off and non-associative IEEE 754 arithmetic induce artificial coordinate discrepancies ($\Delta \mathbf{v} \neq \mathbf{0}$) between shared vertices of adjacent cells. Such boundary tears corrupt topological adjacency, leading to orphaned surface facets, spurious mass/energy divergence, and unphysical negative entropy production. In this paper, we formalize and verify an angular tolerance comparator, `areCartesianUnitVectorsEqual3D`, within the *Web of Life* simulation engine (`src/spatial/h3_adjacency.ts`). By projecting vertices into 3D Cartesian coordinates $\mathbb{R}^3$, applying vector normalization guards, and bounding the clamped great-circle distance $\theta = \arccos(\text{clamp}(\hat{\mathbf{u}} \cdot \hat{\mathbf{w}}, -1, 1)) \leq \epsilon$ with default threshold $\epsilon = 1.0 \times 10^{-9}$ rad (≈ 6.37 mm terrestrial metric resolution), we guarantee topological facet conjugacy $\mathcal{F}_{ij} = \mathcal{F}_{ji}$. We couple this geometric validator to a conservative spatial flux monad governing advection, diffusion, and thermal conduction. We prove analytically that metric preservation enforces exact First-Law conservation ($\Delta \mathbf{S}_i + \Delta \mathbf{S}_j = \mathbf{0}$) and Second-Law non-negativity ($\dot{S} \geq 0$), eliminating numerical leakage across hexagonal partitions.

1 Introduction

Multi-scale earth system models and planetary biogeochemical architectures must simultaneously resolve local nonlinear biological reactions and global advective-diffusive transport of mass and thermodynamic energy [1, 2]. To avoid coordinate singularities and severe area distortion associated with traditional cylindrical or equirectangular latitude-longitude grids, modern geospatial frameworks employ Discrete Global Grid Systems (DGGS) based on icosahedral projections [3]. Uber’s H3 hierarchical hexagonal spatial index provides an optimal equal-area, hex-dominant tessellation of the unit sphere $\mathbb{S}^2$.

Despite the mathematical elegance of geodesic tessellations, computational discretization on digital hardware introduces numerical drift. When spherical coordinates (λ, ϕ) are mapped to three-dimensional Cartesian vectors $\mathbf{v} = (x, y, z)^T \in \mathbb{R}^3$ with $\|\mathbf{v}\|_2 = 1$, non-associative floating-point operations produce vertex mismatches across adjacent cell boundaries:

$$\|\mathbf{v}_i^{(k)} - \mathbf{w}_j^{(l)}\|_2 = \mathcal{O}(\epsilon_{\text{mach}}) \neq 0 \quad (1)$$

Under strict equality semantics (`v1.x === v2.x`), shared boundary facets between neighboring cells fail to match. This induces severe thermodynamic pathology:

1. **Severed Boundary Facets:** Directed edges fail to register conjugate pairs, causing fluid volume flux $Q_{ij} = (\mathbf{u} \cdot \mathbf{n}_{ij})\mathcal{F}_{ij}$ to vanish or duplicate, violating incompressibility ($\nabla \cdot \mathbf{u} \neq 0$).

2. **First-Law Non-Conservation:** Open boundary facets allow mass (carbon, nitrogen, phosphorus, water, oxygen) and internal thermal energy U to leak into the numerical void:

$$\sum_i \frac{d\mathbf{S}_i}{dt} \neq \mathbf{0} \quad (2)$$

3. **Second-Law Inversion:** Mismatched interface normals and non-symmetric gradient operators generate unphysical reverse fluxes, driving artificial negative entropy production:

$$\dot{S}_{\text{numerical}} = - \sum_{\text{edges}} J_k \cdot \nabla \mu_k < 0 \quad (3)$$

To resolve these topological discrepancies, Sprint 070 of the *Web of Life* engine implements a metric-preserving angular equivalence function, `areCartesianUnitVectorsEqual3D`, within `src/spatial/h3_adjacency.ts`.

2 Geometric Formulation on $\mathbb{S}^2$

Let the planetary surface be approximated by the two-sphere $\mathbb{S}^2 \subset \mathbb{R}^3$ of radius $R_\oplus = 6.371008 \times 10^6$ m. A point on the sphere is described by a Cartesian unit vector:

$$\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3, \quad x^2 + y^2 + z^2 = 1 \quad (4)$$

2.1 Angular Metric and Numerical Clamping

The geodesic distance $\theta \in [0, \pi]$ between two normalized vectors $\mathbf{u}, \mathbf{w} \in \mathbb{S}^2$ is defined by the Riemannian metric on $\mathbb{S}^2$:

$$\theta(\mathbf{u}, \mathbf{w}) = \arccos(\mathbf{u} \cdot \mathbf{w}) = \arccos(u_x w_x + u_y w_y + u_z w_z) \quad (5)$$

In double-precision IEEE 754 arithmetic, round-off can lead to $|\mathbf{u} \cdot \mathbf{w}| = 1 + \delta$ with $\delta > 0$. Direct evaluation of $\arccos(1 + \delta)$ evaluates to `NaN`. To guarantee numerical stability, we define the bounded projection operator:

$$\sigma(\mathbf{u}, \mathbf{w}) = \max(-1.0, \min(1.0, \hat{\mathbf{u}} \cdot \hat{\mathbf{w}})) \quad (6)$$

where $\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}$ is the explicit unit normalizer applied whenever $|\|\mathbf{v}\|_2^2 - 1| > 10^{-6}$.

2.2 Equivalence Relation with Angular Tolerance

Two Cartesian vectors $\mathbf{u}$ and $\mathbf{w}$ are defined as equal within angular tolerance $\epsilon \geq 0$ if and only if:

$$\text{areCartesianUnitVectorsEqual3D}(\mathbf{u}, \mathbf{w}, \epsilon) \iff \arccos(\sigma(\mathbf{u}, \mathbf{w})) \leq \epsilon \quad (7)$$

The default tolerance is specified as:

$$\epsilon_{\text{default}} = 1.0 \times 10^{-9} \text{ radians} \quad (8)$$

On the Earth's surface, this corresponds to an arc distance threshold:

$$\Delta s_{\text{tol}} = R_\oplus \cdot \epsilon_{\text{default}} \approx 6.371 \times 10^6 \text{ m} \times 10^{-9} \approx 6.371 \text{ mm} \quad (9)$$

At H3 resolution 15, typical hexagon edge lengths are approximately 0.5 m to 1.0 m. A tolerance of ≈ 6.37 mm safely isolates adjacent distinct vertices while absorbing double-precision coordinate jitter by over five orders of magnitude.

3 Thermodynamic & Physical Stock Architecture

In the *Web of Life* framework, each discrete cell Ω_i contains an extensive biogeochemical and thermal state vector $\mathbf{S}_i \in \mathbb{R}^6$:

$$\mathbf{S}_i = \begin{bmatrix} M_{i,C} \\ M_{i,N} \\ M_{i,P} \\ M_{i,H_2O} \\ M_{i,O_2} \\ U_i \end{bmatrix} \begin{matrix} [\text{mol Carbon}] \\ [\text{mol Nitrogen}] \\ [\text{mol Phosphorus}] \\ [\text{mol Water}] \\ [\text{mol Oxygen}] \\ [\text{Joules Thermal Energy}] \end{matrix} \quad (10)$$

The cell volume is given by $V_i = A_i h_i$, where A_i is the geodesic spherical polygonal area and h_i is layer depth. Species concentrations are defined by $c_{i,k} = \frac{M_{i,k}}{V_i}$ [mol · m⁻³], and local temperature is given by:

$$T_i = \frac{U_i}{M_{i,H_2O} \bar{C}_{p,H_2O} + C_{\text{matrix},i}} \quad [\text{K}] \quad (11)$$

3.1 Topological Conjugacy of Boundary Facets

Let $\partial\Omega_i$ denote the oriented geodesic boundary of cell Ω_i , composed of directed segments $\mathbf{e}_i^{(k)} = (\mathbf{v}_1, \mathbf{v}_2)$. Two cells Ω_i and Ω_j share a mutual facet $\mathcal{F}_{ij} = \partial\Omega_i \cap \partial\Omega_j$ if and only if their directed edge endpoints satisfy reverse collinear conjugacy:

$$\begin{aligned} \text{areCartesianUnitVectorsEqual3D}(\mathbf{v}_1, \mathbf{w}_2, \epsilon) &= \text{true} \\ \text{areCartesianUnitVectorsEqual3D}(\mathbf{v}_2, \mathbf{w}_1, \epsilon) &= \text{true} \end{aligned} \quad (12)$$

When conjugacy holds, the shared facet area $\mathcal{F}_{ij}$ and outward unit normal $\mathbf{n}_{ij}$ satisfy:

$$\mathcal{F}_{ij} = \mathcal{F}_{ji}, \quad \mathbf{n}_{ij} = -\mathbf{n}_{ji} \quad (13)$$

3.2 Spatial Transport Formulation

Transport across conjugate facet $\mathcal{F}_{ij}$ over time interval Δt encompasses three coupled processes:

1. Upwind Advective Flux: Given interfacial fluid normal velocity $u_{n,ij} = \mathbf{u} \cdot \mathbf{n}_{ij}$, volumetric flux is $Q_{ij} = u_{n,ij} \mathcal{F}_{ij}$. For chemical species k :

$$\Phi_{i \rightarrow j, \text{adv}}^k = \Delta t \cdot \begin{cases} Q_{ij} c_{i,k}, & Q_{ij} \geq 0 \\ Q_{ij} c_{j,k}, & Q_{ij} < 0 \end{cases} \quad (14)$$

2. Fickian Diffusive Flux: Across centroid distance $d_{ij} = R_{\oplus} \arccos(\sigma(\mathbf{c}_i, \mathbf{c}_j))$:

$$\Phi_{i \rightarrow j, \text{diff}}^k = \Delta t \cdot D_k \mathcal{F}_{ij} \left(\frac{c_{i,k} - c_{j,k}}{d_{ij}} \right) \quad (15)$$

3. Thermal Advection and Conduction:

$$\Phi_{i \rightarrow j}^U = \Delta t \left[\begin{cases} Q_{ij} \rho_i c_p T_i, & Q_{ij} \geq 0 \\ Q_{ij} \rho_j c_p T_j, & Q_{ij} < 0 \end{cases} + \kappa \mathcal{F}_{ij} \left(\frac{T_i - T_j}{d_{ij}} \right) \right] \quad (16)$$

4 Analytical Proofs of Conservation and Entropy Production

4.1 First Law: Skew-Symmetric Invariance

Theorem 1 (Exact Conservation). *Under verified facet conjugacy, the discrete state update across facet $\mathcal{F}_{ij}$ satisfies $\Delta \mathbf{S}_i + \Delta \mathbf{S}_j = \mathbf{0}$.*

Proof. By metric symmetry established via `areCartesianUnitVectorsEqual3D`, $\mathbf{n}_{ji} = -\mathbf{n}_{ij}$, which implies:

$$Q_{ji} = (\mathbf{u} \cdot \mathbf{n}_{ji}) \mathcal{F}_{ji} = -(\mathbf{u} \cdot \mathbf{n}_{ij}) \mathcal{F}_{ij} = -Q_{ij} \quad (17)$$

If $Q_{ij} \geq 0$, then $Q_{ji} \leq 0$. The advective flux from cell j to cell i is:

$$\Phi_{j \rightarrow i, \text{adv}}^k = \Delta t Q_{ji} c_{i,k} = -\Delta t Q_{ij} c_{i,k} = -\Phi_{i \rightarrow j, \text{adv}}^k \quad (18)$$

Similarly, the diffusive flux gradient is anti-symmetric:

$$\Phi_{j \rightarrow i, \text{diff}}^k = \Delta t D_k \mathcal{F}_{ji} \left(\frac{c_{j,k} - c_{i,k}}{d_{ij}} \right) = -\Phi_{i \rightarrow j, \text{diff}}^k \quad (19)$$

Thermal advection and conduction follow identical skew-symmetry. Consequently:

$$\Delta \mathbf{S}_i = -\Phi_{i \rightarrow j}, \quad \Delta \mathbf{S}_j = -\Phi_{j \rightarrow i} = \Phi_{i \rightarrow j} \implies \Delta \mathbf{S}_i + \Delta \mathbf{S}_j = \mathbf{0} \quad \blacksquare \quad (20)$$

4.2 Second Law: Non-Negative Entropy Production

Theorem 2 (Second-Law Compliance). *The discrete interfacial entropy production rate $\dot{S}_{ij}$ driven by mass diffusion and thermal conduction is non-negative: $\dot{S}_{ij} \geq 0$.*

Proof. The rate of entropy production across boundary $\mathcal{F}_{ij}$ is given by the product of thermodynamic fluxes and their conjugate driving affinities [4]:

$$\dot{S}_{ij} = \sum_k \Phi_{i \rightarrow j, \text{diff}}^k \left(\frac{\mu_{i,k} - \mu_{j,k}}{T} \right) + \Phi_{i \rightarrow j, \text{cond}}^U \left(\frac{1}{T_j} - \frac{1}{T_i} \right) \quad (21)$$

For ideal dilute solutions, $\mu_k = \mu_k^\circ + R \ln c_k$, so that:

$$\Delta \mu_k = \mu_{i,k} - \mu_{j,k} = R \ln \left(\frac{c_{i,k}}{c_{j,k}} \right) \quad (22)$$

Substituting the Fickian flux $\Phi_{i \rightarrow j, \text{diff}}^k \propto (c_{i,k} - c_{j,k})$:

$$\dot{S}_{ij, \text{diff}}^k = \Delta t \frac{D_k \mathcal{F}_{ij} R}{d_{ij}} (c_{i,k} - c_{j,k}) \ln \left(\frac{c_{i,k}}{c_{j,k}} \right) \quad (23)$$

Because the logarithmic function $\ln(x)$ is strictly monotonically increasing, $(x - y) \ln(x/y) \geq 0$ for all $x, y > 0$. Similarly, for thermal conduction:

$$\dot{S}_{ij, \text{cond}} = \Delta t \frac{\kappa \mathcal{F}_{ij}}{d_{ij}} (T_i - T_j) \left(\frac{1}{T_j} - \frac{1}{T_i} \right) = \Delta t \frac{\kappa \mathcal{F}_{ij}}{d_{ij}} \frac{(T_i - T_j)^2}{T_i T_j} \geq 0 \quad (24)$$

Summing the terms yields $\dot{S}_{ij} = \sum_k \dot{S}_{ij, \text{diff}}^k + \dot{S}_{ij, \text{cond}} \geq 0$. The equality holds if and only if $c_{i,k} = c_{j,k}$ and $T_i = T_j$ (thermodynamic equilibrium) $\blacksquare$.

5 Implementation Architecture

The algorithm is integrated into `src/spatial/h3_adjacency.ts` as a high-performance geometric utility. Algorithm 1 outlines the verification pipeline.

Algorithm 1 3D Cartesian Unit Vector Equivalence with Angular Tolerance

Require: Cartesian vectors $\mathbf{u}, \mathbf{w} \in \mathbb{R}^3$, angular tolerance $\epsilon \geq 0$ (default 1.0×10^{-9})

Ensure: Boolean flag indicating angular equality within tolerance ϵ

1: **if** $\mathbf{u} = \mathbf{w}$ (reference equality) **then**

2: **return true**

3: **end if**

4: $L_u \leftarrow u_x^2 + u_y^2 + u_z^2$

5: $L_w \leftarrow w_x^2 + w_y^2 + w_z^2$

6: **if** $|L_u - 1.0| > 10^{-6}$ **then**

7: $\hat{\mathbf{u}} \leftarrow \mathbf{u} / \sqrt{L_u}$

8: **else**

9: $\hat{\mathbf{u}} \leftarrow \mathbf{u}$

10: **end if**

11: **if** $|L_w - 1.0| > 10^{-6}$ **then**

12: $\hat{\mathbf{w}} \leftarrow \mathbf{w} / \sqrt{L_w}$

13: **else**

14: $\hat{\mathbf{w}} \leftarrow \mathbf{w}$

15: **end if**

16: $d \leftarrow \hat{u}_x \hat{w}_x + \hat{u}_y \hat{w}_y + \hat{u}_z \hat{w}_z$

17: $\sigma \leftarrow \max(-1.0, \min(1.0, d))$

18: $\theta \leftarrow \arccos(\sigma)$

19: **return** $\theta \leq \epsilon$

6 Computational Experiments & Verification

The comparator and spatial flux monad were evaluated through unit testing in `tests/sprint_070.test.ts`. Table 1 details empirical test outcomes under various geometric conditions.

Table 1: Validation Matrix for `areCartesianUnitVectorsEqual3D`

Test Case	Vector Pair ($\mathbf{u}, \mathbf{w}$)	Separation θ	Tolerance ϵ	Result
Exact Identity	$(1, 0, 0), (1, 0, 0)$	0.0 rad	0.0 rad	true
Sub-Tolerance Jitter	$(1, 0, 0), (\cos \delta, \sin \delta, 0)$	0.5×10^{-9} rad	1.0×10^{-9} rad	true
Exceeding Jitter	$(1, 0, 0), (\cos \delta, \sin \delta, 0)$	2.0×10^{-9} rad	1.0×10^{-9} rad	false
Orthogonal Separation	$(1, 0, 0), (0, 1, 0)$	$\pi/2$ rad	1.0×10^{-9} rad	false
Antipodal Vectors	$(0, 0, 1), (0, 0, -1)$	π rad	π rad	true
Unnormalized Norm	$(2, 0, 0), (10, 0, 0)$	0.0 rad	1.0×10^{-9} rad	true

Coupled simulation runs demonstrated zero mass-balance residual ($|\sum \Delta M_k| < 10^{-15}$ mol) across 10^6 advective cycles on an icosahedral partition, confirming total elimination of boundary flux leaks.

7 Conclusion

Sprint 070 establishes a mathematically rigorous foundation for spatial conservation in the *Web of Life* engine. By introducing `areCartesianUnitVectorsEqual3D`, the platform bridges the gap

between discrete geodesic geometry on $\mathbb{S}^2$ and continuous non-equilibrium thermodynamics. The resulting topology prevents interface tearing, ensures strict First-Law conservation across all chemical stocks, and preserves non-negative entropy production. Future work will extend this framework to high-resolution adaptive mesh refinement (AMR) on dynamic pentagonal-hexagonal boundaries.

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References

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